MP204 – Electricity & Magnetism

Problem Set 2

Due in the assignment box by 5pm on Friday, 13 March 2015

1. In yet another example of how gravitational and electric fields are very similar, there is a version of Gauss' Theorem for gravity as well:

$$\oint_S \vec{g} \cdot \mathrm{d}\vec{\mathcal{S}} \quad = \quad -4\pi G M_{\mathcal{S}},$$

namely, the surface integral of the gravitational field $\vec{g}$ over a closed surface $\mathcal{S}$ is -4π times Newton's constant G times the mass contained inside $\mathcal{S}$, denoted $M_{\mathcal{S}}$.

Suppose we have a sphere of radius R and total mass M. If the mass density of the sphere is constant, use the above Theorem to find $\vec{g}$ for both $r < R$ and $r > R$.

2. In lecture, we showed that the electric field due to an infinitely long wire of charge was

$$\vec{E} \quad = \quad \frac{\lambda}{2\pi\epsilon_0} \frac{\hat{r}}{r},$$

where λ is the line charge density of the wire, r is the perpendicular distance to the wire and $\hat{r}$ is the unit vector pointing directly away from the wire. Show that if $r > 0$,

$$\vec{\nabla} \cdot \vec{E} \quad = \quad 0$$

and explain why you expected this answer.
Hint: if you'd like to use Cartesian coordinates to do this problem, remember that if the wire lies along the z-axis, then

$$r = \sqrt{x^2 + y^2}, \qquad \hat{r} = \frac{x\hat{\imath} + y\hat{\jmath}}{\sqrt{x^2 + y^2}}.$$

3. The region of space $z < 0$ is filled with a perfect conductor. On the surface of this conductor (the xy-plane), the constant surface charge density is σ. Using Gauss' Theorm, find the electric field everywhere above the plane ($z > 0$).
Big Hint: remember that the electric field in the interior of a perfect conductor is zero.

4. Suppose you have two objects, one of which has charge Q and the other $-Q$. If the potential difference – measured by subtracting the potential of the negatively-charged object from that of the positively-charged one – is $\Delta\Phi$, then the capacitance C of the system is defined to be

$$C = \frac{Q}{\Delta\Phi}.$$

Consider a system of two charged spherical shells with the same centre, where the bigger one has radius a and charge Q and the smaller one has radius b and charge $-Q$. Show that the capacitance of this configuration is

$$C = \frac{4\pi\epsilon_0 ab}{a-b}.$$

(Assume that the charge is uniformly-distributed on each shell.)